

Project Report

ComicCraft - AI Comic Story Creator using Gemini Models

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1. INTRODUCTION

1.1 Project Overview

ComicCraft is a web application that turns a one-line story idea into a finished, illustrated comic. The user enters an idea, a genre and a panel count. Gemini Flash writes a structured outline, a second Gemini model expands it into the full story with a caption and scene prompt for every panel, and Stable Diffusion v1.5 draws each panel. The panels are arranged on pages and exported as a downloadable PDF. The back end is FastAPI, images are composed with Pillow and the PDF is built with fpdf2.

1.2 Purpose

Making a comic needs writing, drawing and layout skills at once. ComicCraft removes that barrier so students, teachers and hobbyists can create and share visual stories without design skills, using only the free Gemini API tier and a local image model.

2. IDEATION PHASE

2.1 Problem Statement

How might we help non-artists turn story ideas into complete illustrated comics quickly, without drawing skills or paid tools? (See Define the Problem Statements.)

2.2 Empathy Map Canvas

The empathy map covers what a student or hobbyist writer thinks, feels, hears, sees, says and does, along with their pains (no drawing skill, slow tools, API limits) and gains (a finished, shareable comic in minutes).

2.3 Brainstorming

The five team members brainstormed ideas, grouped them into clusters and prioritised them by impact and effort. Selected ideas: AI outline, AI story with per-panel prompts, local image generation, automatic layout and PDF export.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The journey follows a creator through Entice, Enter, Engage, Exit and Extend stages: finding the tool, entering an idea, watching the comic being built, downloading the PDF and sharing or regenerating it.

3.2 Solution Requirement

- Functional: idea input, outline generation, story writing, panel image generation, comic layout and preview, PDF export (FR-1 to FR-6).
- Non-functional: usability, security, reliability, performance, availability and scalability (NFR-1 to

NFR-6).

3.3 Data Flow Diagram

The user submits an idea to the FastAPI server; the server calls Gemini Flash for the outline and a second Gemini model for the story, sends each scene prompt to Stable Diffusion, stores the images, builds the layout and returns the PDF. Twelve user stories (USN-1 to USN-12) describe these flows.

3.4 Technology Stack

- Front end: HTML, CSS and JavaScript (Jinja templates)
- Back end: Python and FastAPI
- AI: Gemini Flash (outline), a second Gemini model (story), Stable Diffusion v1.5 with PyTorch (MPS fp16)
- Layout and export: Pillow and fpdf2
- Testing: pytest (24 tests)

4. PROJECT DESIGN

4.1 Problem Solution Fit

Customers: students, teachers and hobbyists who want a quick illustrated story. Their problem is that text tools give no pictures and image tools give no story. ComicCraft fits by producing the story, images and PDF in one flow.

4.2 Proposed Solution

A single web app that covers the whole pipeline, with two Gemini models sharing the free-tier quota and per-panel scene prompts that keep images matched to the story. Business model: freemium with paid plans for more panels, styles and school licences.

4.3 Solution Architecture

Browser -> FastAPI routes -> Gemini Flash, Gemini model 2 and Stable Diffusion -> layout engine -> PDF export -> file storage. See the Solution Architecture document for the diagram.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Work was split into four six-day sprints with 12 user stories and 54 story points: Sprint-1 (10 points, idea input and outline), Sprint-2 (16, story and images), Sprint-3 (15, layout and PDF), Sprint-4 (13, reliability and regeneration). Average velocity is 13.5 points per sprint, or 2.25 points per day.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

- 24 automated pytest tests pass: 7 routes, 7 outline, 1 story, 6 layout, 2 image, 1 PDF.
- About 12.4 s per panel image and about 104 s end to end on Apple M5 (16 GB, MPS fp16).
- Gemini rate-limit (429) errors are caught and reported; two models share the quota (5 requests per minute, 20 per day per model on the free tier).

7. RESULTS

7.1 Output Screenshots

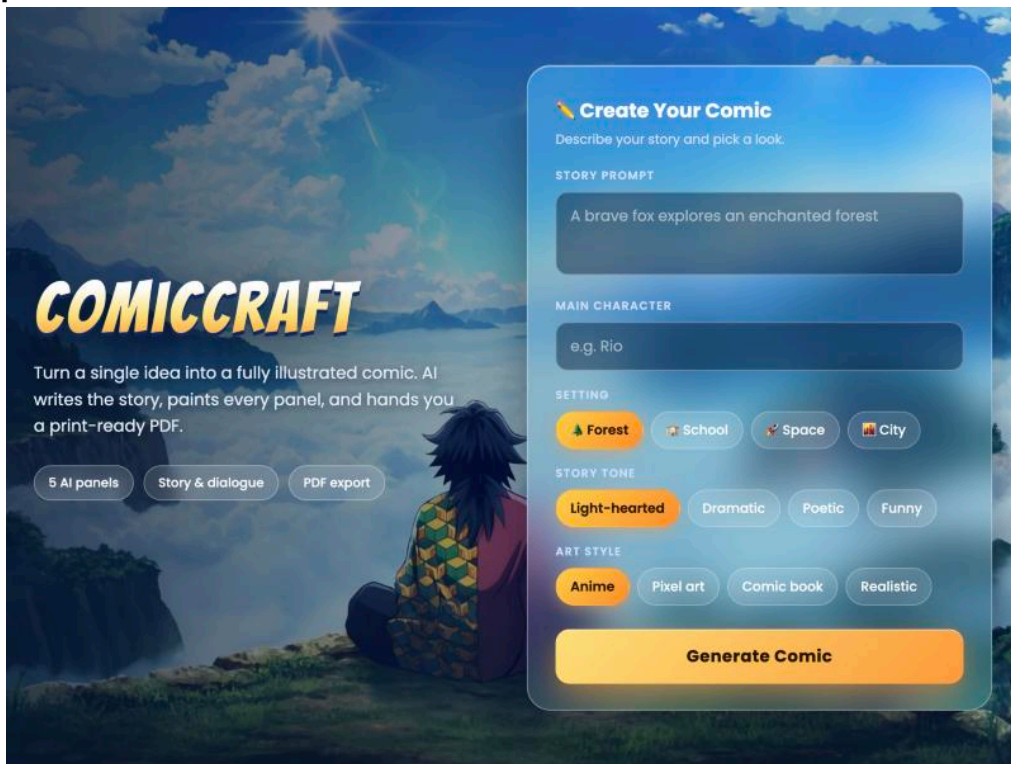


Figure 1: Home page

Your Comic

Panel 1: The Threshold of Adventure



Figure 2: Generated comic preview



Figure 3: A generated panel

Panel 1: The Threshold of Adventure



Rio, a small orange fox wearing an oversized green scarf and steampunk-style goggles, stands confidently at the entrance of the Whispering Woods. The trees are tall with shimmering silver leaves, and a path of soft moss leads into the depths of the vibrant greenery.

Caption: The threshold of the Whispering Woods awaits.

Rio adjusted his goggles and wrapped his oversized green scarf tighter against the morning chill. "Today is the day," he chirped to the silver trees, his tail twitching with anticipation. He took a bold step forward, ready to uncover the secrets hidden within the glowing canopy.

Figure 4: A page of the exported PDF

8. ADVANTAGES & DISADVANTAGES

Advantages

- One tool for outline, story, images, layout and PDF
- Works on the free Gemini tier and a local image model
- No drawing or design skill needed
- Fast: a full comic in about 104 s

Disadvantages

- Image generation needs a capable laptop or GPU
- Free Gemini quota limits how many comics can be made per day
- Outline is capped at 5 panels
- Image style and character consistency depend on the Stable Diffusion model

9. CONCLUSION

ComicCraft shows that a story idea can be turned into a complete illustrated comic PDF automatically by combining two Gemini models, Stable Diffusion and a simple layout engine. The system meets its functional requirements and all 24 automated tests pass.

10. FUTURE SCOPE

- Consistent characters across panels
- More art styles and page layouts
- User accounts and a gallery of saved comics
- Cloud GPU deployment with a job queue
- Speech balloons and multi-language stories

11. APPENDIX

- Source Code: <https://github.com/Chippiiiiiii/comic-craft>
- Dataset Link: not applicable (pre-trained models are used; no dataset was collected)
- GitHub & Project Demo Link: <https://github.com/Chippiiiiiii/comic-craft>